Stop the Fire!

A Network Optimization Approach to Wildfire Containment

IE 492 — Senior Design Project · Midterm Presentation

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NP-Hard

Graph Theory

Firefighter Problem

April 2026

MOTIVATION

Why Wildfire Containment Matters

Environmental
Millions of hectares of forests destroyed annually worldwide

Human Cost
Thousands displaced, lives lost in uncontrolled wildfires

Economic Damage
Billions in property damage and firefighting costs each year

Reactive → Proactive
Current response is reactive — we need strategic decision-making

"Current response is reactive — we need proactive, strategic decision-making."

Framing: Pitching to OGM (General Directorate of Forestry)

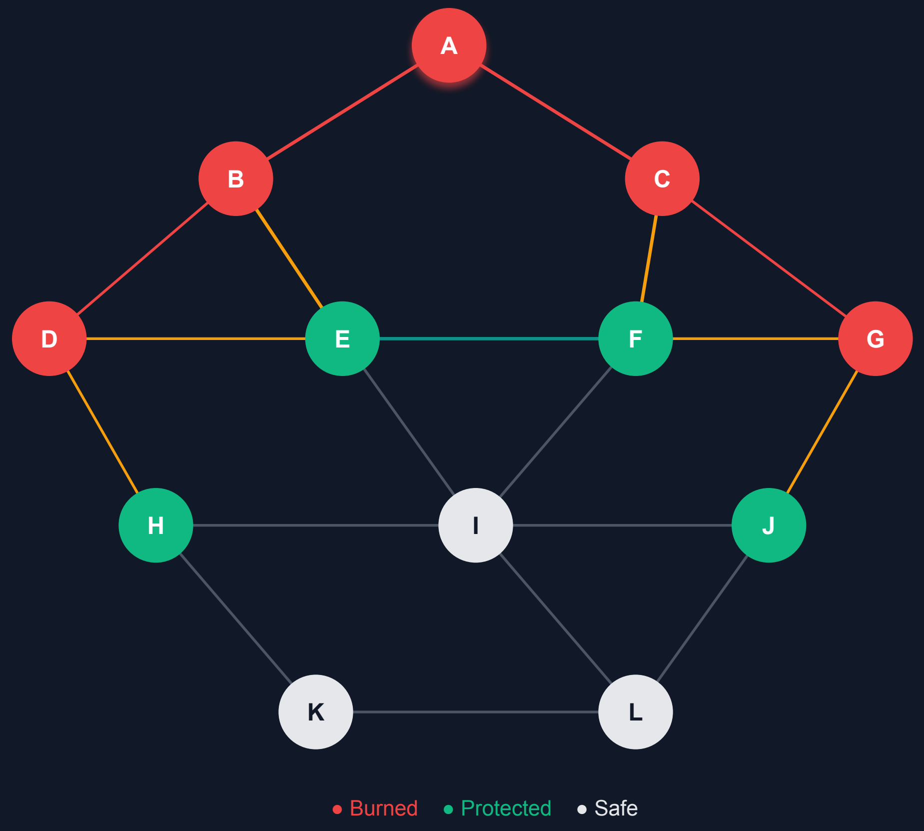
CRITERION 1: PROBLEM

The Firefighter Problem

FORMAL STATEMENT

"Given a graph G, a fire origin, and k resources per turn, minimize total burned nodes."

- Classical combinatorial optimization problem
- **NP-Hard** — no polynomial-time exact solution
- Deterministic network optimization with limited intervention resources
- Our adaptation: planar/map-based graphs simulating real geography



T=0 Fire ignites at A
T=1 Protect D → B, C burn
T=2 Protect G → E, F burn | H–L remain safe

CRITERION 1: PROBLEM

Our Project Scope

OBJECTIVE

Comparatively analyze which intervention strategy (**Greedy** vs. **Min-Cut** vs. **Hybrid**) minimizes **K (total burned nodes)**

Real Map → **Network Graph** → **Strategy Application** → **Performance Comparison**

KEY SCOPE DECISIONS

- Primary Goal: Minimize K (total burned nodes), not just resource efficiency
- Different graph structures (density, connectivity) favor different strategies

CRITERION 2: REQUIREMENTS

Model Assumptions

Discrete & Uniform Propagation

Time moves in discrete steps; fire spreads to all unprotected neighbors simultaneously

Irreversibility of Damage

Burned nodes cannot be recovered — prevention only, no extinguishing

Absolute Immunity

Protected nodes are permanently safe; fire cannot pass through them

Full Observability

Algorithm has perfect information of the entire network at all times

Note: Nodes whose all neighbors are protected are also considered safe.

CRITERION 2: REQUIREMENTS

Model Constraints

Dynamic Resource Constraint
k = 2 nodes protected per turn (scalable parameter based on graph density)

Topological Constraints
Planar graphs; edges = spatial adjacency; natural & unnatural barriers = no edge

Binary State
Nodes are strictly: Unburned, Burned, or Protected — no partial damage

Static Environment
No wind, elevation, or flammability variation in baseline model

Computational Complexity
NP-Hard; min-cut computation scales exponentially with graph size

CRITERION 2: REQUIREMENTS

Limitations & Future Extensions

MODEL LIMITATIONS

Deliberate decisions — not shortcomings

Binary Node States

Unburned · Burned · Protected

Fine vertex partitioning keeps real error margin low

Deterministic Model

No wind · No terrain slope · No fuel moisture

Fixed rules only — no randomness in baseline

n = 30 Vertices

Synthetic planar graph — controlled baseline

Minimum required for statistically meaningful results

Static Graph Topology

Edges and connectivity fixed throughout run

Graph structure does not change mid-simulation

PLANNED EXTENSIONS

Staged additions after core validation

Stochastic Spread

Wind direction · Elevation · Flammability coeff.

Fire propagation becomes asymmetric and realistic

Scalable Graph Size

Vertex count can be increased for larger scenarios

Protection budget k scales with graph density

Partial Burning + Timed Intervention

Edge-based propagation · time-based discrete steps

Gradual fire damage + scheduled resource deployment

Real Map — Ege Region

scipy.spatial.Delaunay → real geographic network

OGM risk map integration as final phase

CRITERION 3: DESIGN PROCESS

Formal Design Process

STAGE 01

Problem Formulation

Graph model G=(V,E)

Fire mechanics & rules

Constraints defined

Objective: minimize total burned

STAGE 02

Network Generation

Random 2D point set

Delaunay triangulation

Planar graph G=(V,E)

`scipy.spatial.Delaunay`

Export to NetworkX

STAGE 03

Strategy Design

Greedy: Max Degree

Greedy: Max White Nbrs

Min-Cut Edge / Vertex

Hybrid (if-else logic)

STAGE 04

Simulation Engine

Each turn: protect → spread

Protect top-k WHITE nodes

Fire spreads simultaneously

Stop: no WHITE near RED

STAGE 05

Comparative Analysis

Same graph, 4 strategies

Same start point, vary graph

Different topologies tested

Metric: fewest burned zones

CRITERION 3: DESIGN PROCESS

Simulation Model & Mechanics

1 **Strategy selects priority zones**

Ranks fire-front zones: by degree · neighbors · min-cut

2 **Engine protects top 1 zones (k = 1)**

Permanently immune — fire cannot pass through them

3 **Fire spreads simultaneously**

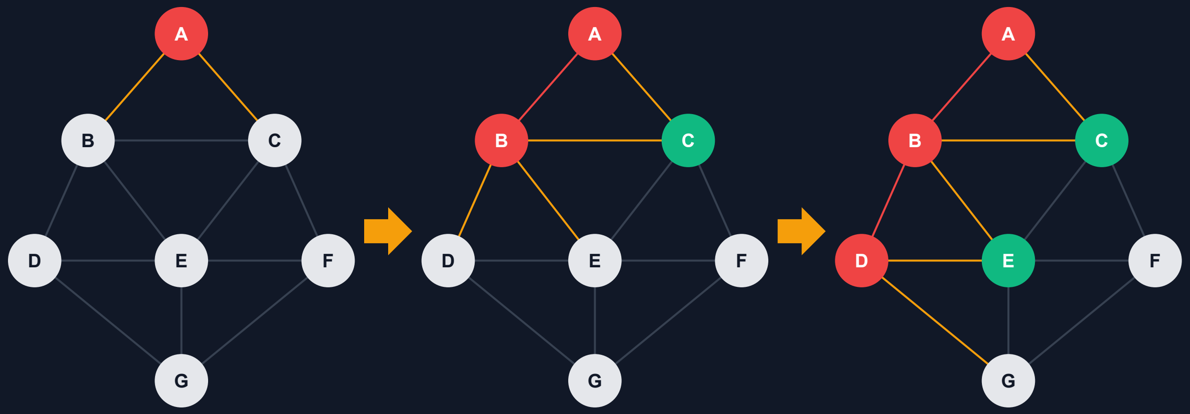
Every burning zone ignites all adjacent unprotected zones at once

4 **Repeat until contained**

Stop when no unburned zone borders a burning zone

GRAPH STRUCTURE

Connected planar graph, default n = 30
RED = burning | GREEN = protected | WHITE = safe
Stops when no WHITE vertex is adjacent to RED



T = 0 (Fire starts) T = 1 (Protect C, B burns) T = 2 (Protect D, E burns)

• Burned • Protected • Safe — Fire Front

Fire Front: $F = \{ v \mid state(v) = WHITE \wedge \exists u \in N(v) : state(u) = RED \}$

WHAT IS THE FIRE FRONT?

Fire Front = every unburned zone that borders at least one burning zone

This is the only set the strategy can act on each turn — protect now or lose it.

Unburned (safe) · Burning · Protected (immune) | 2 zones protected per turn (k = 2)

Strategies: Greedy Baselines

Strategy 1 — Max Degree

Pick: Fire-front vertex with highest degree

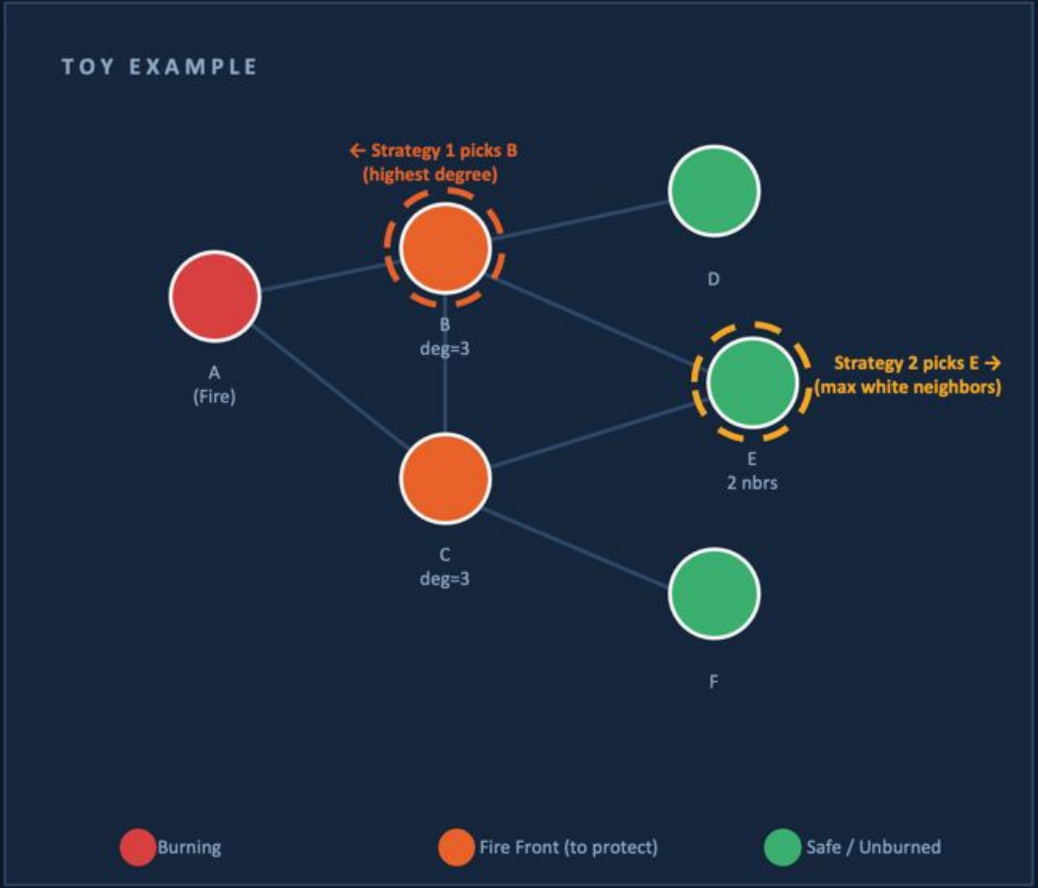
- ✓ Blocks high-connectivity hubs
- ✗ May miss low-degree narrow separators

Strategy 2 — Max White Neighbors

Pick: Vertex touching most unburned territory

- ✓ Preserves larger safe zones
- ✗ Still myopic — one-step lookahead only

Both strategies are ***local*** — they react to current state without seeing the global graph.



Strategies: Min-Cut Approaches

Strategy 3 — Min-Cut Edge Front

1. Build augmented graph: source → RED → WHITE → sink
2. Compute minimum edge cut
3. Protect WHITE endpoints on the fire front

≈ ***Finds firewall areas*** (indirect — we protect vertices, not edges)

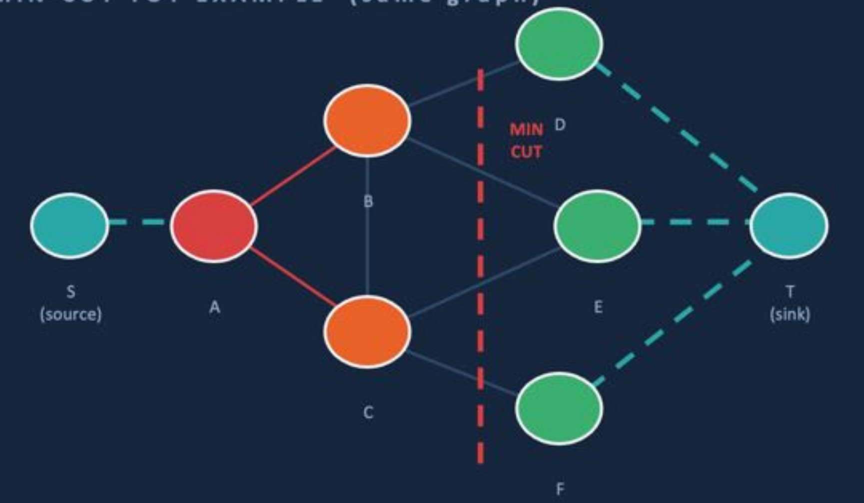
Strategy 4 — Min-Cut Vertex Front

1. Identify distant valuable WHITE targets
2. Compute minimum node cut: RED → targets
3. Prioritize cut vertices on fire front

✓ **Directly aligned** with vertex-protection mechanics

Hybrid idea: Min-Cut early → switch to Greedy when connectivity is high.

MIN-CUT TOY EXAMPLE (same graph)



INPUT

Augmented graph
(RED set + WHITE set + source/sink)

OUTPUT

Bottleneck edges
→ protect WHITE endpoints on fire front

"Min-Cut tells us WHERE the bottleneck is"

CRITERION 3: DESIGN PROCESS

Test Methodology & Design

TEST CONFIGURATION

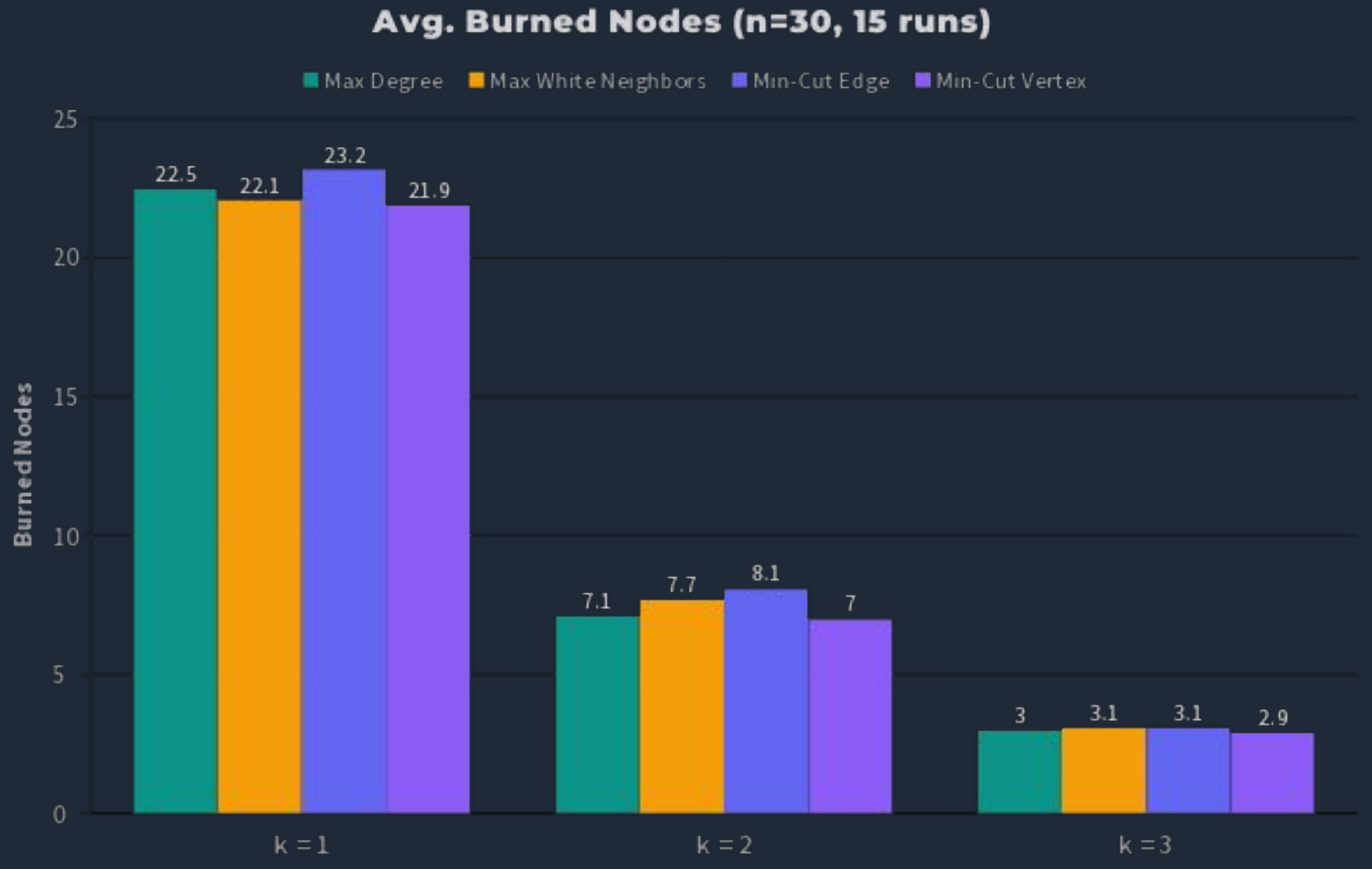
Graph Sizes
n = 20, 30, 40 vertices

Resources per Turn
k = 1, 2, 3

Runs per Config
15 randomized runs, averaged

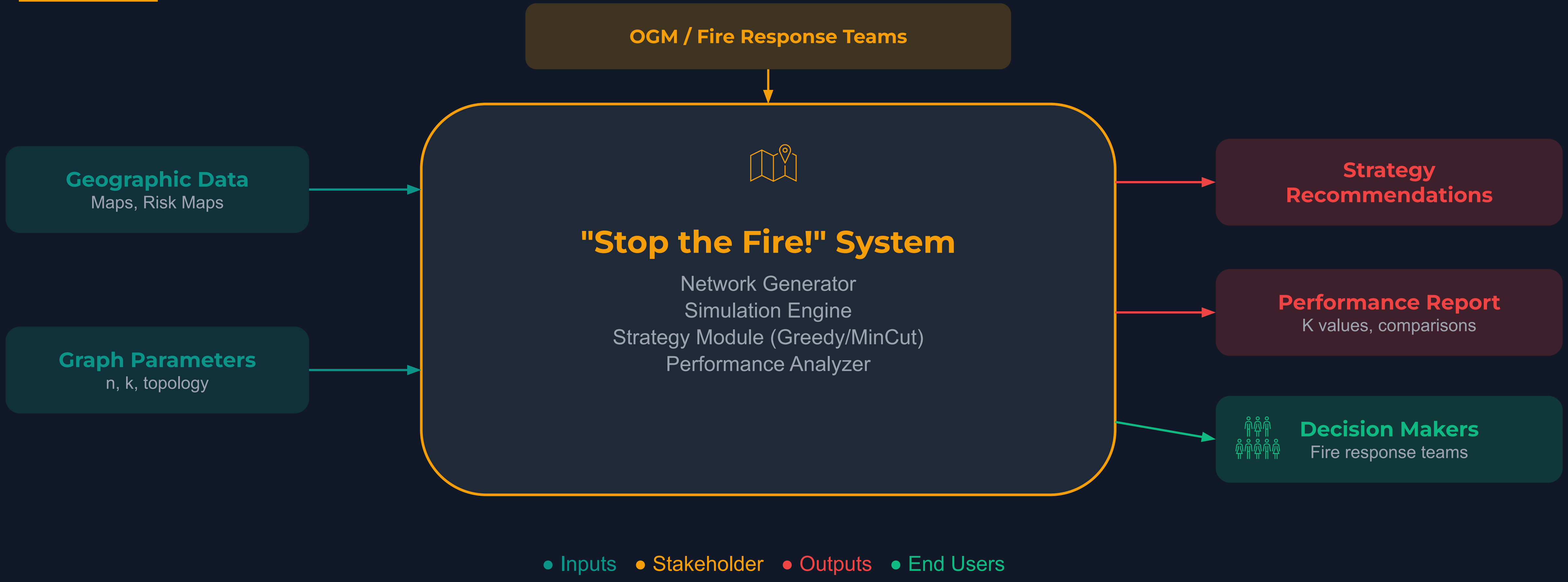
Strategies Tested
Max Degree
Max White Neighbors
Min-Cut Edge
Min-Cut Vertex

Primary Metric
Burned nodes (K) — lower is better



CRITERION 4: SYSTEMIC VIEW

Context Diagram



Current Progress & Next Steps

✓ COMPLETED

- ✓ Problem formulation & mathematical model
- ✓ Simulation engine with turn-based mechanics
- ✓ 4 strategies implemented
- ✓ Toy example validation

🔄 IN PROGRESS

- → Hybrid strategy development (If-Else switching logic)
- → Comprehensive testing with varied parameters

⬜ NEXT PHASE

- ○ Real map-to-network conversion
- ○ Stochastic extensions (wind, terrain)
- ○ Large-scale performance benchmarking

PROJECT STATUS: ~60% COMPLETE

Core engine working · Strategies validated · Testing in progress · On track for final delivery

Conclusion & Key Takeaways

1

The Firefighter Problem provides a rigorous framework for wildfire containment optimization

2

Min-Cut strategies show promise for strategic bottleneck identification

3

No single strategy dominates — topology matters → hybrid approaches needed

4

Our system serves as a proactive decision-support tool, not just a reactive response

Thank you — Questions?

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